

TOT

Assignment-1

by

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Architectuse

OB TOT

Layer-1 Peg-ception layer

- Sensor: Collected data from

the environmental

are

Pestroom actions.

- devices: Microcontroller, system-on-chip (SoC) and other hardware components.

Communication protocol: bluetooth, Zigbee, Z-wave

Layer-2

Network layer

Connectivity: wib,
thesnel, Cellulas network
(2G/3G/4G/5G)

Network protocol: IPv6, IPV4
COAP

MQTT, etc.

Data Bransmission: Device- to Device

(D2D) , Device to cloud
(D2C)

Network

Te

Notisk Topologies: Star,
mesh, Tree, etc.

Layer 3

middle layer

es

Data Processing: Real time dataprocessing

biomwage updates, and device catool.

Device Manayonet : Clould storage detachages

and data warehousing

Data Storage : APIS, SDIC; and

Lots

developement

Application Enablement:

Remote monitering, biomage

update.

Security authentication, authentication, authorization

encryption and access
Control.

Layers-4

Application layers

- IoT application: Smart homes,

Industrial IoT

automation

health care etc

- Data analytics: Predictive

analytics

machine learning and data

visualization user interface: web,

mobile, voice based

interface

Business

logic

:

Logic Integration with

Building business workflow as

budget

business processes.

Additional

Components.

searity: Seave boot, seare
firmware updates
and feare data storage.

двожда

- Power Management: Energy
hajuafting, Paver

Grisuption optimization
and bally
scalability Horizontal
Scaling, vertical

scaling

Enty posability: Entegration
with different
devices, protocols with system,

IOT architectise Palloon

Device cloud device (CD) Devices
with could, which processes
Communicate

late and

Sends
Conmarl

back
to back

Device -Device (DD):
Devices

directly with each other
on the cloud

Device Gateway-should
(DGC)

Devices
Communicate with
a

local gateway, which then

sinds date to

Communicate **without** relying

ould

IOT Architective challenges

secoity: Ensures the Secinotty
of

device, data and
Communication Protocal

Scalability: Scaling IT
system to

Member of

accommodate large number
devices and data

interoperability: Integrating
devices and

System from different manufacturers
and protocols.

Power Management:
Managing Power

Consumption
and optimizing Cres
energy

efficiency in IoT
devices.